

implicit Euler step



continuous gradient flow





















































W









Distance from different distribution



Function variant to minimize

Same as euclidean gradient flows!

JKO Scheme

- One **implicit Euler step** in Wasserstein space:

$$\rho_{t+1} = \arg \min_{\rho} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_t) \right\}$$

Function we want to minimize

Distance from current distribution

- Time-step parameter $\tau > 0$.

- Balance between:

Same as euclidean gradient flows!

- lowering energy $\mathcal{F}(\rho)$
- staying close to previous iterate (in Wasserstein distance)

- Small τ : W_2 weighs more, stay close to current ρ_t , smaller step size
- As $\tau \rightarrow 0$, the discrete scheme $\rightarrow$ **continuous gradient flow**.

Historical Context & Importance of JKO

- Jordan–Kinderlehrer–Otto (1998):



Richard Jordan



David Kinderlehrer



Felix Otto

SIAM J. MATH. ANAL.
Vol. 29, No. 1, pp. 1–17, January 1998

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THE VARIATIONAL FORMULATION OF THE FOKKER–PLANCK EQUATION*

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In memory of Richard Duffin

Abstract. The Fokker–Planck equation, or forward Kolmogorov equation, describes the evolution of the probability density for a stochastic process associated with an Ito stochastic differential equation. It pertains to a wide variety of time-dependent systems in which randomness plays a role. In this paper, we are concerned with Fokker–Planck equations for which the drift term is given by the gradient of a potential. For a broad class of potentials, we construct a time discrete, iterative variational scheme whose solutions converge to the solution of the Fokker–Planck equation. The major novelty of this iterative scheme is that the time-step is governed by the Wasserstein metric on probability measures. This formulation enables us to reveal an appealing, and previously unexplored, relationship between the Fokker–Planck equation and the associated free energy functional. Namely, we demonstrate that the dynamics may be regarded as a gradient flux, or a steepest descent, for the free energy with respect to the Wasserstein metric.

Key words. Fokker–Planck equation, steepest descent, free energy, Wasserstein metric